

Statistical Efficiency of Birth-Death Tree Parameter Estimation

Stat 700

Santosh Desai, Matt McAnear, Urvi Mehta

Department of Statistics, University of Michigan

December 4, 2025

The Phylodeep Package

Phylodeep Paper is by J. Voznica et al [1]

- ▶ Parameter estimation is difficult and MLE method can be challenging for larger trees.
- ▶ Major components of the paper to address this
 - ▶ CBLV representation of trees
 - ▶ Pre-trained neural networks for pre-specified tree types
- ▶ The goal is to speed up inference

Usage

- ▶ Input: Phylogenetic tree in Newick format
- ▶ Output: Estimated parameters (e.g., birth rate (λ), death rate (μ), R_0)
- ▶ Given a tree, Phylodeep selects the pre-trained model with the highest probability
- ▶ Outputs actually come with uncertainty estimates from a parametric bootstrap

Tree Generation

- ▶ Trees were generated in parallel using the 'treesimulator' package
- ▶ Explored Phylopomp for this, but format is slightly different and packages weren't quickly interoperable

```
1 # Generate a BD tree with 200-500 tips ,  
2 # transmission rate of 0.5, removal rate  
3 # of 0.25 and sampling probability of 0.3  
4 generate_bd --min_tips 200 --max_tips 500 \  
5   --la 0.5 --psi 0.25 --p 0.3 \  
6   --nwk tree.nwk --log params.csv
```

Maximum Likelihood Estimation: Overview

- ▶ **Goal:** Estimate birth rate (λ) and death rate (μ) from phylogenetic trees
- ▶ **Method:** Numerical optimization of Stadler (2010) likelihood
- ▶ **Key Feature:** Works with **tree data only** - no nucleotide sequences required
- ▶ **Implementation:** Uses `scipy.optimize.minimize` with L-BFGS-B algorithm

Advantages:

- ▶ Statistically principled (maximum likelihood)
- ▶ Works on small trees (no minimum tip size requirement)
- ▶ Provides point estimates with optimization diagnostics

Stadler (2010) Likelihood Formulation

For a constant-rate birth-death process with sampling probability ρ :

$$\begin{aligned}\ell(\lambda, \mu | T, \rho) = & (n - 1) \log(\lambda) \\ & - (\lambda + \mu) T_{\text{total}} \\ & + n \log(\rho) \\ & - r \cdot T + \text{penalty terms}\end{aligned}$$

where:

- ▶ n = number of tips
- ▶ T_{total} = total branch length
- ▶ T = tree height (time from root to present)
- ▶ $r = \lambda - \mu$ = net diversification rate
- ▶ ρ = sampling probability

Key insight: Survival probability term ($-r \cdot T$) helps identify μ separately from λ

MLE Implementation: Tree Statistics Extraction

Extracted from each tree:

- ▶ Number of tips: n
- ▶ Total branch length: $T_{\text{total}} = \sum_{e \in E} \ell(e)$
- ▶ Tree height: $T = \max_{v \in V} d(\text{root}, v)$
- ▶ Branching times: All internal node ages
- ▶ Mean branch length: $\bar{\ell}$

Initial parameter estimates (method-of-moments):

$$\hat{r} = \frac{\log(n)}{T}$$

$$\hat{\lambda} = \frac{n-1}{T_{\text{total}}}$$

$$\hat{\mu} = \max(0.01, \hat{\lambda} - \hat{r})$$

These serve as starting values for numerical optimization.

MLE Implementation: Optimization

Numerical optimization:

- ▶ **Algorithm:** L-BFGS-B (bounded optimization)
- ▶ **Bounds:** $\lambda \in [0.01, 10.0]$, $\mu \in [0.01, 10.0]$
- ▶ **Constraint:** $\mu < \lambda$ (enforced via penalty)
- ▶ **Gradient:** Analytical gradient provided for faster convergence

Gradient components:

$$\frac{\partial \ell}{\partial \lambda} = -\frac{n-1}{\lambda} + T_{\text{total}} + T$$
$$\frac{\partial \ell}{\partial \mu} = T_{\text{total}} - T + \text{penalty term}$$

Regularization: Penalty term prevents μ from collapsing to lower bound in large trees

MLE Implementation: Key Features

1. Survival Probability Term

- ▶ Includes $-r \cdot T$ term to help identify μ
- ▶ Accounts for probability of observing n tips given λ and μ
- ▶ Critical for separating μ from λ in the likelihood

2. Regularization for Large Trees

- ▶ Penalty: $-0.1 \log(\mu)$ when $\mu < 0.1$ and $n > 10$
- ▶ Prevents μ from getting stuck at lower bound (0.01)
- ▶ Encourages realistic death rate estimates based on tree size

3. Robust Error Handling

- ▶ Checks for invalid trees ($n \leq 1$, $T_{\text{total}} \leq 0$)
- ▶ Handles optimization failures gracefully
- ▶ Returns success status, log-likelihood, and iteration count

4. Method-of-Moments Initialization

- ▶ Smart starting values: $\hat{r} = \log(n)/T$, $\hat{\lambda} = (n-1)/T_{\text{total}}$
- ▶ Ensures faster convergence and avoids local minima

MLE: Advantages and Limitations

Advantages:

- ▶ Statistically principled (maximum likelihood)
- ▶ No minimum tree size requirement
- ▶ Works on small trees ($n < 50$)
- ▶ Provides optimization diagnostics
- ▶ Fast computation (seconds per tree)
- ▶ Based on well-established Stadler (2010) theory

Limitations:

- ▶ Higher RMSE than PhyloDeep on large trees
- ▶ Point estimates only (no uncertainty quantification)
- ▶ Requires careful initialization for convergence
- ▶ Can struggle with very large trees ($n > 500$)

Best use case: Small to medium trees where statistical rigor is important

Phylodeep Methodology

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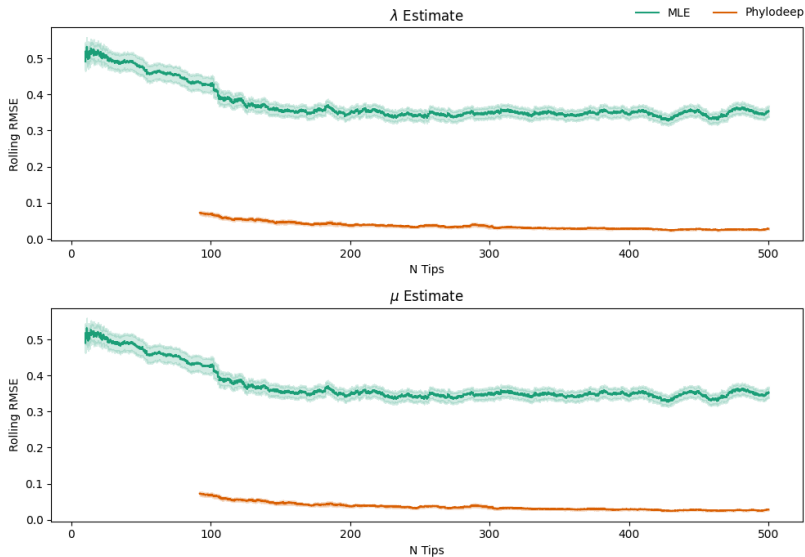


Figure: Rolling bootstrapped RMSE estimates of μ and λ by tree size and resulting 95% CI. Window size is 500.

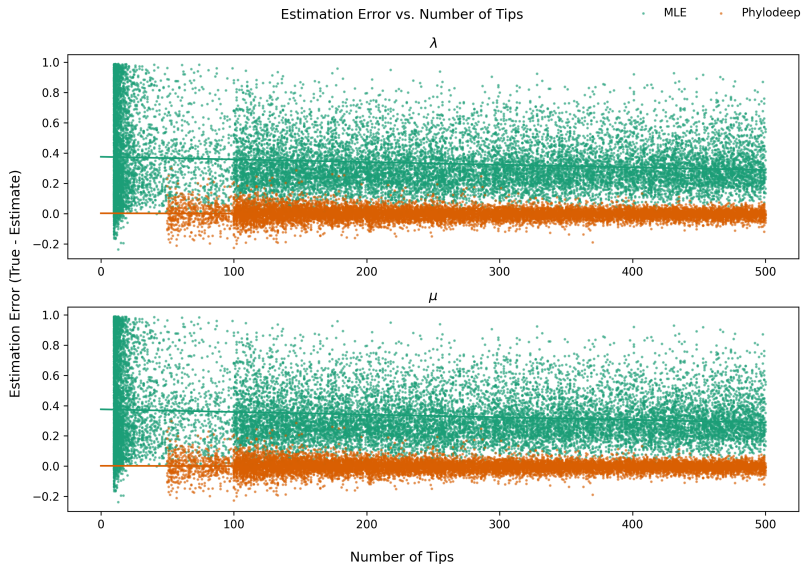


Figure: Raw plot of the absolute errors in λ, μ by tree size.

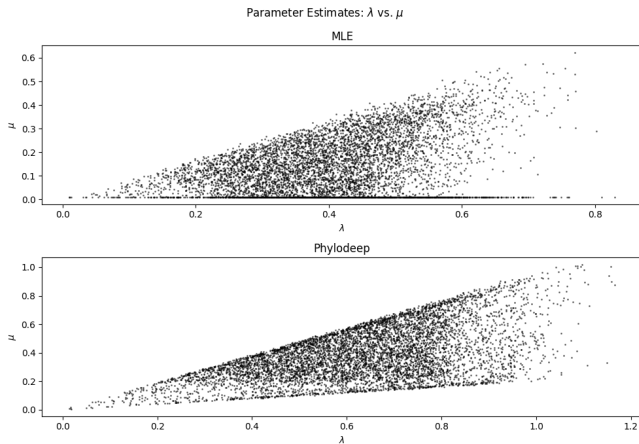


Figure: Plot of λ vs μ estimates to show distribution of estimates.

Main takeaways:

- ▶ RMSE decreases with increasing tree size for both methods.
- ▶ MLE can be used on smaller trees - Phylodeep requires a minimum tip size of around 50.
- ▶ Otherwise, Phylodeep has lower RMSE than MLE, probably due to being trained on 3.9M trees.

References

- [1] J. Voznica et al. “Deep learning from phylogenies to uncover the epidemiological dynamics of outbreaks”. en. In: *Nature Communications* 13.1 (July 2022). Publisher: Nature Publishing Group, p. 3896. ISSN: 2041-1723. DOI: 10.1038/s41467-022-31511-0. URL: <https://www.nature.com/articles/s41467-022-31511-0> (visited on 11/04/2025).